

Assignment

What is the use of following libraries:

1.requests

2.bs4

Ans:

1. requests Library

The **requests** library is used to send HTTP requests to websites and retrieve their content. It allows Python programs to communicate with web servers and download web pages, APIs, and other online resources.

Uses of **requests**:

- Send HTTP requests (GET, POST, etc.).
- Download HTML content from websites.
- Access REST APIs and retrieve data.
- Upload data to web servers.
- Handle headers, cookies, and authentication.

Example:

```
import requests

url = "https://example.com"
response = requests.get(url)

print(response.status_code)
print(response.text)
```

2. bs4 (BeautifulSoup) Library

bs4 (BeautifulSoup) is a Python library used for parsing HTML and XML documents. It helps extract specific information from web pages by navigating and searching the HTML structure.

Uses of **BeautifulSoup**:

- Parse HTML and XML documents.
- Extract text, links, images, tables, and other webpage elements.
- Search elements using tags, classes, or IDs.
- Clean and organize scraped web data.
- Commonly used with the `requests` library for web scraping.

Example:

```
import requests
from bs4 import BeautifulSoup

url = "https://example.com"
response = requests.get(url)

soup = BeautifulSoup(response.text, "html.parser")

print(soup.title.text)
```

Difference between **requests** and **bs4**

requests	bs4 (BeautifulSoup)
Used to fetch web pages from the internet.	Used to parse and extract data from HTML/XML.
Sends HTTP requests to a server.	Processes the downloaded HTML content.
Retrieves the webpage source code.	Finds and extracts specific elements like headings, links, and tables.
Works before parsing.	Works after the webpage has been downloaded.